# Real-Time Pressure Data Pipeline for Wind Tunnel Experiments

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**Project:** Optical Vibration Testing Project
**Internship:** Summer Research Internship

## 1. Project Summary and Expected Results

During my internship, I developed a real-time data pipeline for wind tunnel experiments using a Scanivalve MPS4200 pressure scanner. The aim was to transfer measurements to a computer, process them automatically and store them in a database for later analysis. The system was designed to provide reliable data transmission, accurate timestamps, easy retrieval and a setup that could be reproduced by future users.

## 2. Setup / Prerequisites

The system used a data-acquisition PC, a BeagleBone Black for Precision Time Protocol (PTP) synchronisation, a Scanivalve MPS4200 and an isolated local network. MATLAB, Telegraf and InfluxDB 3.0 were used for simulation, data processing and storage. The Scanivalve transmitted binary data using UDP, which Telegraf parsed before sending it to InfluxDB.

An Excel configuration file stored database settings, network parameters and sensor metadata. This separated experiment-specific settings from the MATLAB code and made the system easier to reconfigure.

## 3. Progress Made and Key Results

I first configured PTP synchronisation and UDP communication between the system components. Wireshark was used to verify that measurement packets reached the acquisition computer.

I then developed a MATLAB simulation to test the pipeline without relying entirely on the physical scanner. The simulation generated test points, added small variations to represent measurement data, converted them into binary UDP packets and sent them to Telegraf. This allowed the complete data path to be tested in a controlled environment.

I automated the Telegraf configuration using sensor definitions stored in Excel. This allowed incoming binary packets to be parsed according to the configured sensor channels and stored in InfluxDB 3.0.

Finally, I developed a MATLAB query interface that retrieves data by time range or run ID, loads the configured sensor channels and exports results to Excel. The simulation also retrieves stored data, calculates averages for each test point and produces structured run records.

I also produced a setup guide covering synchronisation, UDP configuration, packet verification, Telegraf, database output and troubleshooting. Together, these tools provide an end-to-end workflow from measurement acquisition to data retrieval.

## 4. Troubleshooting and Error Handling

Testing identified several practical issues. If Telegraf reports “bind: address already in use”, another process is using the UDP port; stopping that process and restarting Telegraf resolves the problem. Data may also be temporarily unavailable after streaming while it is being processed, so the MATLAB workflow retries the database query before retrieving the available data. If the Excel sensor configuration is missing or invalid, the configuration must be corrected, with default channels available in the query interface if necessary.

The current system assumes that the network, UDP port, database credentials and sensor information are correctly configured. It does not yet automatically recover from network failures, invalid packets, prolonged database outages or missing data. These areas should be improved before long periods of unattended data collection.

## 5. Next Steps

The next step is to test the pipeline with longer experiments and higher data rates to assess reliability and identify packet loss or ingestion problems. Further improvements could include stronger validation and recovery features and real-time visualisation of pressure measurements.

## 6. Positive Impact and What I Learnt

The project reduces manual data handling and provides a more consistent way to transfer, store and retrieve experimental measurements. The automated configuration and documentation also make the system easier for future users to adapt and troubleshoot.

Through the internship, I gained practical experience in network communication, time synchronisation, binary data, real-time data pipelines, database querying and MATLAB automation. I also learnt how to break a complex system into smaller components and test them separately.

One of the main lessons was that a working prototype is not enough in a research environment. A system also needs to be understandable, repeatable and robust so that others can use it.

## 7. Reflection on My Interest in Postgraduate Research

The internship gave me a clearer understanding of postgraduate research in a laboratory environment. I enjoyed solving a practical problem where computing supported physical experiments, and I learnt that research involves repeated testing, troubleshooting and refinement.

The experience increased my appreciation of postgraduate research while also showing me that my main interest is in practical applications of computing and developing systems that solve real-world problems. This will help me consider postgraduate study and the balance between research and practical technical work.